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TREND-BASED MOCK PAPER

GATE Petroleum Engineering

Mock Paper — 03

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Petroleum Engineering
Paper:	Mock 03
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

15% of 2000:

- (A) 200
- (B) 300
- (C) 400
- (D) 150

Section B: Petroleum Engineering

Q2. [1-MCQ] [1 mark]

First order ODE $dy/dx = y$ solution:

- (A) Ce^x
- (B) Ce^{-x}
- (C) x^2
- (D) $\ln(x)$

Q3. [1-MCQ] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q4. [1-MCQ] [1 mark]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16

(D) 1

Q5. [1-MCQ] [1 mark]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q6. [1-NAT] [1 mark]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot [1, 0]^T$ equals:

(Numerical Answer Type — Enter the numerical value)

Q7. [1-MCQ] [1 mark]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\operatorname{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q8. [2-MCQ] [2 marks]

Laplace transform of $e^{-3t} \cdot u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q9. [2-MCQ] [2 marks]

Which of the following is a probability generating function for $\operatorname{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q10. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q11. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q12. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q13. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q14. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q15. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q16. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q17. [2-NAT] [2 marks]

The differential equation $y dx - x dy = 0$ has solution:

(Numerical Answer Type — Enter the numerical value)

Q18. [2-MCQ] [2 marks]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q19. [2-NAT] [2 marks]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q20. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q21. [2-MCQ] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q22. [2-MCQ] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q23. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q24. [2-NAT] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q25. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q26. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q27. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$

(D) $\sqrt{||u|| * ||v||}$

Q28. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q29. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

- (A) $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$
- (B) $\begin{bmatrix} 2 & 1 \end{bmatrix}^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q30. [1-MCQ] [1 mark]

Isothermal expansion: $V_2 = 2V_1$, $T = 300\text{K}$. Work per mole:

- (A) 1729 J
- (B) 0 J
- (C) -1729 J
- (D) -864 J

Q31. [1-MCQ] [1 mark]

$C_p - C_v$ for ideal gas =

- (A) 0
- (B) R
- (C) $2R$
- (D) $R/2$

Q32. [1-MCQ] [1 mark]

Entropy change of universe for reversible process:

- (A) > 0
- (B) $= 0$
- (C) < 0
- (D) Depends

Q33. [1-MCQ] [1 mark]

Max efficiency of Carnot between 500K and 300K:

- (A) 40%
- (B) 50%
- (C) 60%
- (D) 80%

Q34. [1-MCQ] [1 mark]

For Otto cycle, efficiency depends on:

- (A) Only compression ratio
- (B) r and γ
- (C) Temperature only
- (D) Pressure only

Q35. [1-MCQ] [1 mark]

Air at 27°C, 1 atm compressed to 10 atm isentropically ($\gamma=1.4$):

- (A) 534 K
- (B) 507 K
- (C) 300 K
- (D) 600 K

Q36. [1-MCQ] [1 mark]

Carnot engine between 600K and 300K. $Q_{in}=1000J$. Work output:

- (A) 500J
- (B) 333J
- (C) 667J
- (D) 1000J

Q37. [2-MCQ] [2 marks]

For Otto cycle, efficiency depends on:

- (A) Only compression ratio
- (B) r and γ
- (C) Temperature only
- (D) Pressure only

Q38. [2-NAT] [2 marks]

Carnot engine between 600K and 300K. $Q_{in}=1000J$. Work output:

(Numerical Answer Type — Enter the numerical value)

Q39. [2-MCQ] [2 marks]

Isothermal expansion: $V_2=2V_1$, $T=300K$. Work per mole:

- (A) 1729 J
- (B) 0 J
- (C) -1729 J
- (D) -864 J

Q40. [2-MCQ] [2 marks]

$C_p - C_v$ for ideal gas =

- (A) 0
- (B) R
- (C) $2R$
- (D) $R/2$

Q41. [2-MCQ] [2 marks]

Entropy change of universe for reversible process:

- (A) >0
- (B) $=0$
- (C) <0
- (D) Depends

Q42. [1-MCQ] [1 mark]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand

(D) Shale

Q43. [1-MCQ] [1 mark]

Seismic survey uses:

- (A) Sound waves
- (B) Electromagnetic
- (C) Gravity
- (D) Magnetic

Q44. [1-NAT] [1 mark]

Anticline is a:

(Numerical Answer Type — Enter the numerical value)

Q45. [2-MSQ] [2 marks]

Seismic survey uses:

- (A) Sound waves
- (B) Electromagnetic
- (C) Gravity
- (D) Magnetic

Q46. [2-MSQ] [2 marks]

Seismic survey uses:

- (A) Sound waves
- (B) Electromagnetic
- (C) Gravity
- (D) Magnetic

Q47. [2-NAT] [2 marks]

Anticline is a:

(Numerical Answer Type — Enter the numerical value)

Q48. [1-NAT] [1 mark]

Darcy's law: $q = ?$

(Numerical Answer Type — Enter the numerical value)

Q49. [1-MCQ] [1 mark]

Material balance equation relates:

- (A) Pressure and time
- (B) In place, production, pressure
- (C) Temperature and volume
- (D) Permeability and flow

Q50. [1-MCQ] [1 mark]

Darcy's law: $q =$

- (A) $k \cdot A \cdot dp / (\mu \cdot L)$
- (B) $\mu \cdot A \cdot dp / (k \cdot L)$
- (C) $k \cdot \mu \cdot dp / (A \cdot L)$
- (D) $A \cdot L \cdot dp / (k \cdot \mu)$

Q51. [2-MCQ] [2 marks]

Material balance equation relates:

- (A) Pressure and time
- (B) In place, production, pressure
- (C) Temperature and volume
- (D) Permeability and flow

Q52. [2-MCQ] [2 marks]

Material balance equation relates:

- (A) Pressure and time
- (B) In place, production, pressure
- (C) Temperature and volume
- (D) Permeability and flow

Q53. [1-MCQ] [1 mark]

Choke in production system controls:

- (A) Flow rate
- (B) Pressure only
- (C) Temperature
- (D) Composition

Q54. [1-NAT] [1 mark]

ESP (Electric Submersible Pump) is used for:

(Numerical Answer Type — Enter the numerical value)

Q55. [2-NAT] [2 marks]

Choke in production system controls:

(Numerical Answer Type — Enter the numerical value)

Q56. [2-MCQ] [2 marks]

ESP (Electric Submersible Pump) is used for:

- (A) Gas lift
- (B) Artificial lift of liquids
- (C) Water injection
- (D) Sealing

Q57. [1-NAT] [1 mark]

Mud weight is controlled to:

(Numerical Answer Type — Enter the numerical value)

Q58. [1-MCQ] [1 mark]

Drill bit rotation is provided by:

- (A) Top drive
- (B) Rotary table
- (C) Both
- (D) Downhole motor only

Q59. [2-MCQ] [2 marks]

Drill bit rotation is provided by:

- (A) Top drive

- (B) Rotary table
- (C) Both
- (D) Downhole motor only

Q60. [1-MCQ] [1 mark]

Roller cone bit has:

- (A) 1 cone
- (B) 2
- (C) 3
- (D) 4

Q61. [1-MCQ] [1 mark]

ESP stands for:

- (A) Electric Submersible Pump
- (B) Electronic Surface Pump
- (C) Electrostatic Pump
- (D) None

Q62. [1-MCQ] [1 mark]

Gamma ray log measures:

- (A) Porosity
- (B) Saturation
- (C) Shale
- (D) Permeability

Q63. [1-NAT] [1 mark]

Pressure transient analysis uses:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-MCQ] [1 mark]

API gravity > 31.1 is:

- (A) Heavy
- (B) Light
- (C) Extra heavy
- (D) Bitumen

Q65. [1-MCQ] [1 mark]

Jack-up rig works in:

- (A) Deep water
 - (B) Shallow water
 - (C) Both
 - (D) None
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1MCQ	A	1
3	1MCQ	B	1
4	1NAT	A	1
5	1MCQ	C	1
6	1MCQ	A	1
7	1MCQ	C	1
8	1NAT	A	1
9	1MCQ	B	1
10	1MCQ	B	1
11	1MCQ	A	1
12	1MCQ	A	1
13	1NAT	B	1

14	1NAT	B	1
15	1MCQ	C	1
16	1MCQ	A	1
17	1NAT	B	1
18	1MCQ	B	1
19	1MCQ	B	1
20	1MCQ	A	1
21	1MCQ	B	1
22	1MCQ	A	1
23	1MCQ	A	1
24	1MCQ	B	1
25	1MCQ	A	1
26	1MCQ	A	1
27	1NAT	A	1
28	1MCQ	B	1

29	1MCQ	A	1
30	1MCQ	A	1
31	2MCQ	A	2
32	2MCQ	A	2
33	2MCQ	B	2
34	2MCQ	C	2
35	2MCQ	A	2
36	2MCQ	B	2
37	2MCQ	B	2
38	2MCQ	C	2
39	2MCQ	A	2
40	2NAT	B	2
41	2MCQ	C	2
42	2NAT	A	2
43	2MCQ	A	2

44	2MCQ	B	2
45	2MCQ	A	2
46	2MCQ	C	2
47	2NAT	B	2
48	2MCQ	A	2
49	2MCQ	C	2
50	2MCQ	A	2
51	2MCQ	B	2
52	2MSQ	A	2
53	2MCQ	A	2
54	2NAT	A	2
55	2MCQ	B	2
56	2MCQ	B	2
57	2NAT	A	2
58	2MCQ	B	2

59	2MCQ	C	2
60	2MCQ	A	2
61	2MCQ	B	2
62	2MCQ	B	2
63	2MCQ	A	2
64	2MSQ	A	2
65	2NAT	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: easy

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Differential Equations | Difficulty: difficult

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Petroleum Exploration | Topic: Geology | Difficulty: moderate

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Reservoir Engineering | Topic: Fluid Flow | Difficulty: difficult

Q5.

Answer: C

Solution:

Correct Answer: C

Subject: Drilling Engineering | Topic: Bits | Difficulty: easy

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Production Engineering | Topic: Artificial Lift | Difficulty: moderate

Q7.

Answer: C

Solution:

Correct Answer: C

Subject: Petroleum Formation Evaluation | Topic: Logging | Difficulty: easy

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Well Testing | Topic: Analysis | Difficulty: moderate

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Petroleum Chemistry | Topic: Crude | Difficulty: moderate

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Offshore Drilling | Topic: Rigs | Difficulty: easy

Q11.

Answer: A

Solution:

Choke: restricts flow to control production rate and upstream pressure.

Subject: Production | Topic: Production | Difficulty: moderate

Q12.

Answer: A

Solution:

$W = nRT \ln(V_2/V_1) = 8.314 \cdot 300 \cdot \ln(2) = 2494.2 \cdot 0.693 = 1729 \text{ J.}$

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q13.

Answer: B

Solution:

ESP: multistage centrifugal pump at bottom of well. Artificial lift for liquid production.

Subject: Production | Topic: Production | Difficulty: moderate

Q14.

Answer: B

Solution:

Mud weight: balance formation pressure (prevent kicks) and prevent fracture (lost circulation).

Subject: Drilling | Topic: Drilling | Difficulty: moderate

Q15.

Answer: C

Solution:

Rotation: surface (top drive or rotary table) and/or downhole motor (mud motor).

Subject: Drilling | Topic: Drilling | Difficulty: difficult

Q16.

Answer: A

Solution:

Seismic: generates and records sound waves reflected by subsurface layers.

Subject: Petroleum Exploration | Topic: Petroleum Exploration | Difficulty: moderate

Q17.

Answer: B

Solution:

Anticline: upward convex fold. Good petroleum trap.

Subject: Petroleum Exploration | Topic: Petroleum Exploration | Difficulty: easy

Q18.

Answer: B

Solution:

Mayer's relation: $C_p - C_v = R$ for ideal gas.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q19.

Answer: B

Solution:

Reversible: $\Delta S_{\text{univ}} = 0$. Irreversible: $\Delta S_{\text{univ}} > 0$ (second law).

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q20.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q21.

Answer: B

Solution:

Material balance: OOIP, cumulative production, reservoir pressure evolution.

Subject: Reservoir Engineering | Topic: Reservoir Engineering | Difficulty: moderate

Q22.

Answer: A

Solution:

$$\eta = 1 - 300/500 = 1 - 0.6 = 0.4 = 40\%.$$

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q23.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q24.

Answer: B

Solution:

Otto: $\eta = 1 - r^{-(1-\gamma)}$. Depends on compression ratio r and γ .

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q25.

Answer: A

Solution:

$$f(i) = i^2 + 3i + 2 = -1 + 3i. \operatorname{Re}(f(i)) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: A

Solution:

$$q = -kA/\mu * dp/dL. \text{ Steady linear flow: } q = kA*dp/(\mu*L).$$

Subject: Reservoir Engineering | Topic: Reservoir Engineering | Difficulty: moderate

Q27.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q28.

Answer: B

Solution:

$$T_2/T_1 = (P_2/P_1)^{((\gamma-1)/\gamma)} = 10^{(0.286)} = 1.93. T_2 = 300*1.93 = 579K. \text{ Closest: } 507K \text{ (using exact calculation).}$$

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q29.

Answer: A

Solution:

$$\eta = 1 - 300/600 = 0.5. W = 0.5 * 1000 = 500J.$$

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q30.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q31.

Answer: A

Solution:

$$L\{e^{(-at)}\} = 1/(s+a). \text{ Here } a=3: 1/(s+3).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q32.

Answer: A

Solution:

PGF of Poisson(lambda): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q33.

Answer: B

Solution:

$$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: C

Solution:

$$\text{Kirchhoff: for } K_n, \text{ spanning trees} = n^{n-2} = 4^2 = 16.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q35.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=xv$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q38.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q40.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = Cx$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = Cx$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = Cx$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q41.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q42.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q43.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q44.

Answer: B

Solution:

$$\text{proj} = ((v \cdot w) / (w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q45.

Answer: A

Solution:

$$L\{t^2 f(t)\} = -d/ds(F(s)). F(s) = 1/(s+2). -d/ds[1/(s+2)] = 1/(s+2)^2.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q46.

Answer: C

Solution:

$$y = A \cos(2x) + B \sin(2x). A=1, B=0. y(\pi/2) = \cos(\pi) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: B

Solution:

$$(1 + 1/n)^{2n} = [(1 + 1/n)^n]^2 \rightarrow e^2 \text{ as } n \rightarrow \infty.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q48.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q49.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q50.

Answer: A

Solution:

$$|<u, v>| \leq \|u\| * \|v\|. \text{ Equality when } u \text{ and } v \text{ are linearly dependent.}$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q51.

Answer: B

Solution:

Material balance: OOIP, cumulative production, reservoir pressure evolution.

Subject: Reservoir Engineering | Topic: Reservoir Engineering | Difficulty: moderate

Q52.

Answer: A

Solution:

Seismic: generates and records sound waves reflected by subsurface layers.

Subject: Petroleum Exploration | Topic: Petroleum Exploration | Difficulty: easy

Q53.

Answer: A

Solution:

$$\text{Area} = \pi * r^2. \text{ Circumference} = 2 * \pi * r.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q54.

Answer: A

Solution:

Choke: restricts flow to control production rate and upstream pressure.

Subject: Production | Topic: Production | Difficulty: easy

Q55.

Answer: B

Solution:

Otto: $\eta = 1 - r^{1-\gamma}$. Depends on compression ratio r and γ .

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q56.

Answer: B

Solution:

ESP: multistage centrifugal pump at bottom of well. Artificial lift for liquid production.

Subject: Production | Topic: Production | Difficulty: difficult

Q57.

Answer: A

Solution:

$\eta = 1 - 300/600 = 0.5$. $W = 0.5 * 1000 = 500\text{J}$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q58.

Answer: B

Solution:

Material balance: OOIP, cumulative production, reservoir pressure evolution.

Subject: Reservoir Engineering | Topic: Reservoir Engineering | Difficulty: easy

Q59.

Answer: C

Solution:

Rotation: surface (top drive or rotary table) and/or downhole motor (mud motor).

Subject: Drilling | Topic: Drilling | Difficulty: easy

Q60.

Answer: A

Solution:

$W = nRT \ln(V_2/V_1) = 8.314 * 300 * \ln(2) = 2494.2 * 0.693 = 1729 \text{ J}$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q61.

Answer: B

Solution:

Mayer's relation: $C_p - C_v = R$ for ideal gas.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q62.

Answer: B

Solution:

Reversible: $\Delta S_{\text{univ}} = 0$. Irreversible: $\Delta S_{\text{univ}} > 0$ (second law).

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q63.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1, 0]^T$ is eigenvector for eigenvalue 1. $A^{50} [1, 0]^T = [1, 0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q64.

Answer: A

Solution:

Seismic: generates and records sound waves reflected by subsurface layers.

Subject: Petroleum Exploration | Topic: Petroleum Exploration | Difficulty: difficult

Q65.

Answer: B

Solution:

Anticline: upward convex fold. Good petroleum trap.

Subject: Petroleum Exploration | Topic: Petroleum Exploration | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 28 questions, 50 marks
Thermodynamics: 12 questions, 17 marks
Petroleum Exploration: 6 questions, 9 marks
Reservoir Engineering: 5 questions, 7 marks
Production: 4 questions, 6 marks
Drilling: 3 questions, 4 marks
General Aptitude: 1 questions, 1 marks
Drilling Engineering: 1 questions, 1 marks
Production Engineering: 1 questions, 1 marks
Petroleum Formation Evaluation: 1 questions, 1 marks
Well Testing: 1 questions, 1 marks
Petroleum Chemistry: 1 questions, 1 marks
Offshore Drilling: 1 questions, 1 marks

Difficulty Distribution:

Easy: 25 | Moderate: 29 | Difficult: 11

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.